

1 2.1 Generalized Procrustes Analysis (GPA)

Problem: Training shapes have unknown rigid transformations. Need aligned landmarks for PCA.

Solution: Iteratively estimate similarity transforms:

1. **Translation:** $X_i \leftarrow X_i - \bar{X}_i$
2. **Scale:** $\|X_i\|_F = 1$
3. **Rotation:** $\min_R \|X_i - R\mu\|_F$, $\text{SVD}(X_i^\top \mu = U\Sigma V^\top)$, $R = UV^\top$
4. Update $\mu = \frac{1}{M} \sum R_i X_i$, repeat until convergence

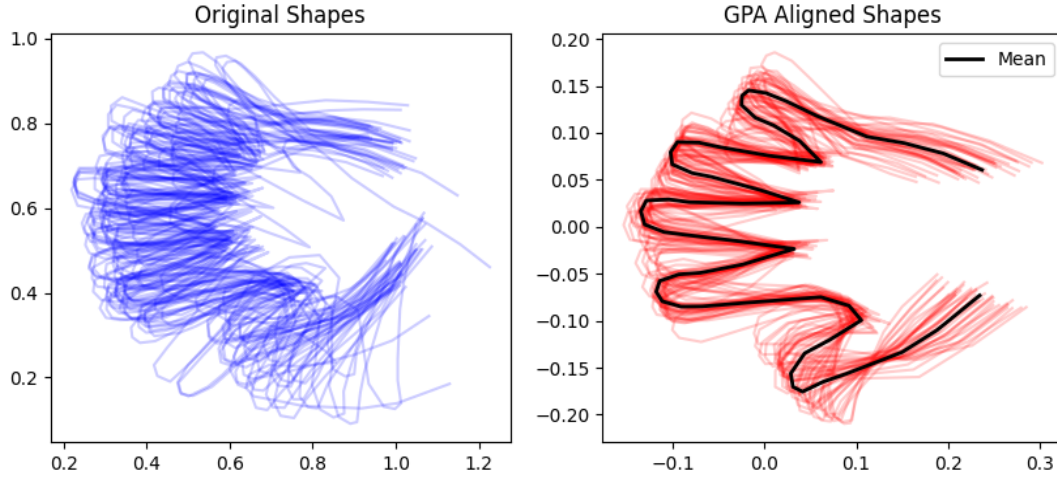


Figure 1: Original (left) vs GPA-aligned shapes (right) with mean (black).

2 2.2 Statistical Shape Model

2.1 2.2.1 PCA Subspace Model

Aligned landmarks $w \in \mathbb{R}^{112}$. Covariance $C = WW^\top / (M - 1)$. Eig: $C = V\Lambda V^\top$.

Model: $\hat{w} = \mu + V_N b$, $N = 3$ ($\geq 90\%$ energy).

2.2 2.2.2 Probabilistic PCA

Generative: $w = Wz + \epsilon$, $z \sim \mathcal{N}(0, I)$, $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$.

Learning: $\sigma^2 = 1.798 \times 10^{-5}$, $W = V_N(\Lambda_N - \sigma^2 I)^{1/2}$.

3 2.3 Inference

Test aligned via Procrustes, then:

- **PCA:** $b = V_N^\top(w - \mu)$, $\hat{w} = \mu + V_N b$
- **PPCA:** $z = (W^\top W + \sigma^2 I)^{-1} W^\top(w - \mu)$, $\hat{w} = \mu + W z$

Results:

- $N = 3$, $\sigma^2 = 1.7978 \times 10^{-5}$
- PCA MSE: 2.212337×10^{-5}
- PPCA MSE: 2.212435×10^{-5}

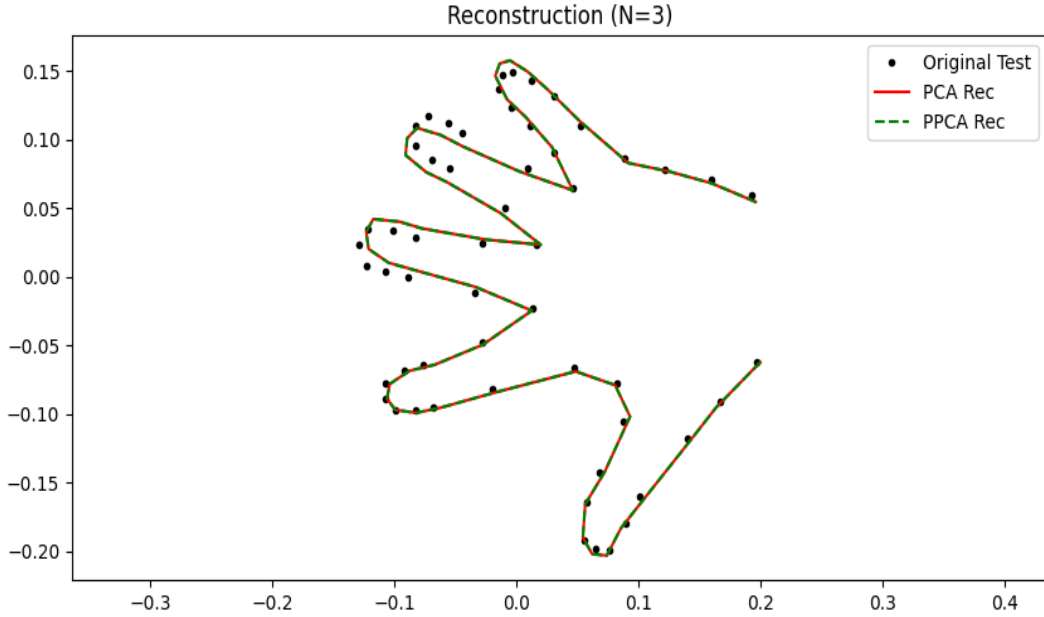


Figure 2: Reconstruction: test (black), PCA (red), PPCA (green).

4 2.4 Discussion

GPA initialization: Robust. SVD rotations are closed-form optima; converges independently of initial μ .

PCA vs PPCA: MSEs nearly identical ($\sigma^2 \ll \lambda_{1:3}$). PPCA shrinks coefficients via noise model, theoretically better generalization.